

„ZPR PWr – Zintegrowany Program Rozwoju Politechniki Wrocławskiej”

Game level design

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https://pwr-edu.zoom.us/rec/share/W_9XAZ80yKPGeUSSWJFgMGt96lih5d2J-Pep5ZOxxQa4y4hij_yzunH1B_WEadGU.KnQ8W-snIJ10otvb

Lecture

1. **Silniki gier. Wprowadzenie do Unity. Pierwsza gra 2D.**
2. **Mechanika 2D, animacje.**
3. **Oświetlenie, tekstury, materiały. Pierwsza gra 3D.**
4. Narzędzia wspierające, np. Blender.
5. **Zapis/odczyt danych, komunikacja sieciowa.**
6. **Modelowanie terenu. Generowanie terenu.**
7. **Sztuczna inteligencja w grach.**
8. Virtual Reality, wsparcie VR w Unity.
9. **Optymalizacje w Unity.**
10. Historia i klasyfikacja gier.
11. Prototypowanie gier. GDD.
12. Projektowanie poziomów gier.
13. Projektowanie gier dla różnych platform.
14. Testowanie gier. Test.
15. Test.

Game Development

Ideas

- Concept Art
- Gameplay concept
- Story concept
- Gimmick

"designed primarily to attract attention or increase appeal, often with little intrinsic value"

https://www.reddit.com/r/Games/comments/7vlryh/where_does_a_gimmick_end_and_a_legitimate_feature/

Execution

- Balancing difficulty and fairness
- Pacing
- **Level Design**
- Controls
- Physics
change direction in mid-air ?
double jump ?

[1] "Introduction to Level Design" by Rico Lemba

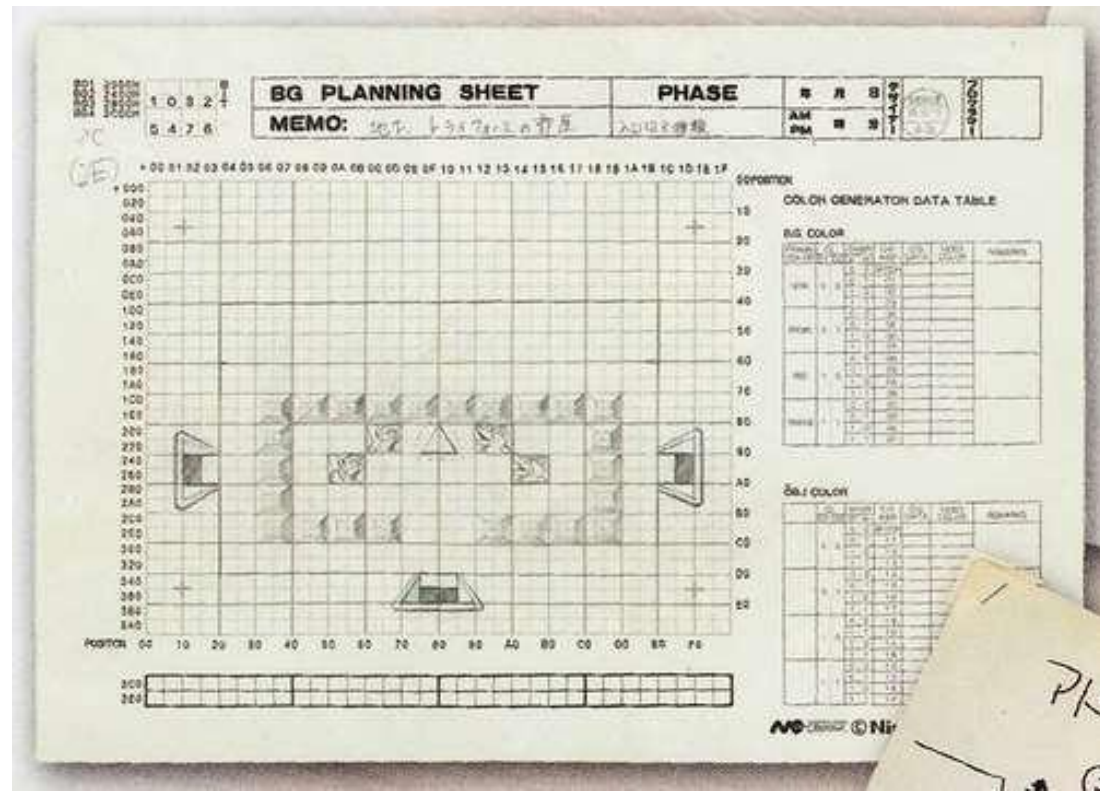
<https://www.slideshare.net/RicoLemba/introduction-to-level-design-49277472>

Level Design

- Theme / Level Concept
- Enemy Design
- Obstacle Design
- Puzzle Design
- Enemy Placement
- Obstacle Placement

<https://www.gamingalexandria.com/wp/2019/07/legend-of-zelda-development-files/>

not environment design!



Level Design

WHY IS IT IMPORTANT?

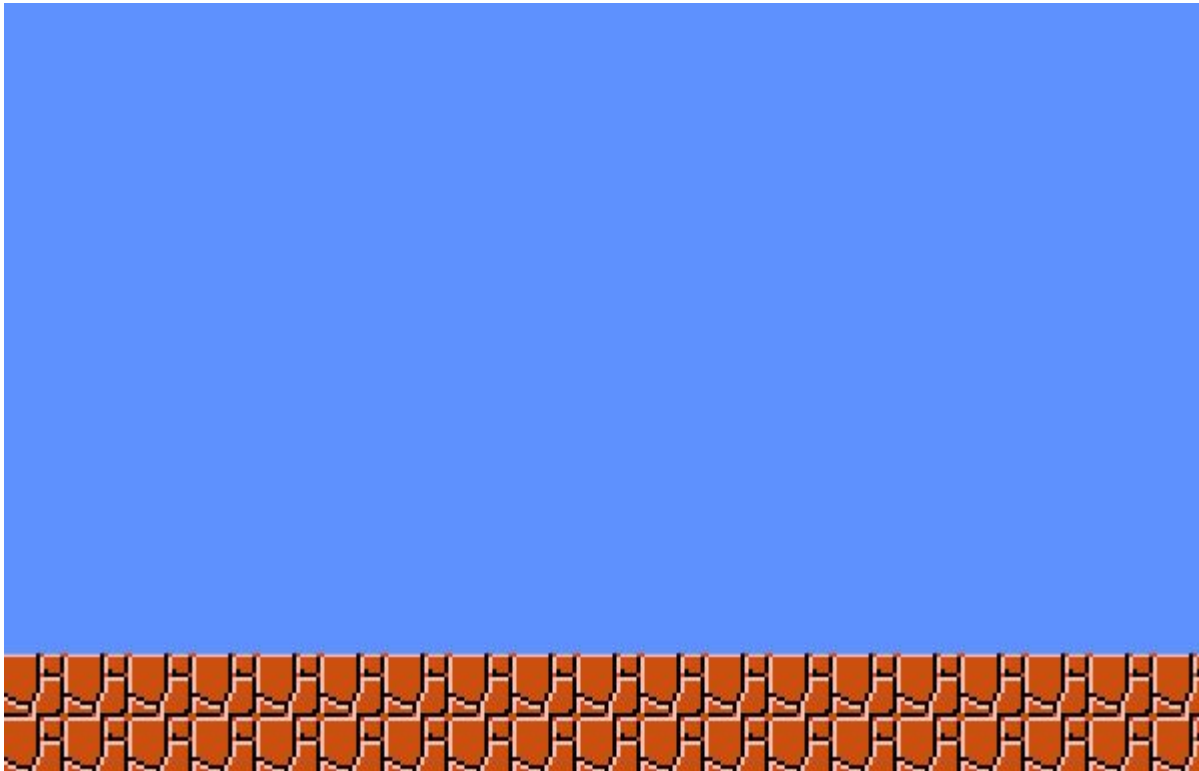
Provides Useful Mastery

- Mario's Abilities
 - Jump
 - Destroy blocks
 - Run
 - Shoot fireballs



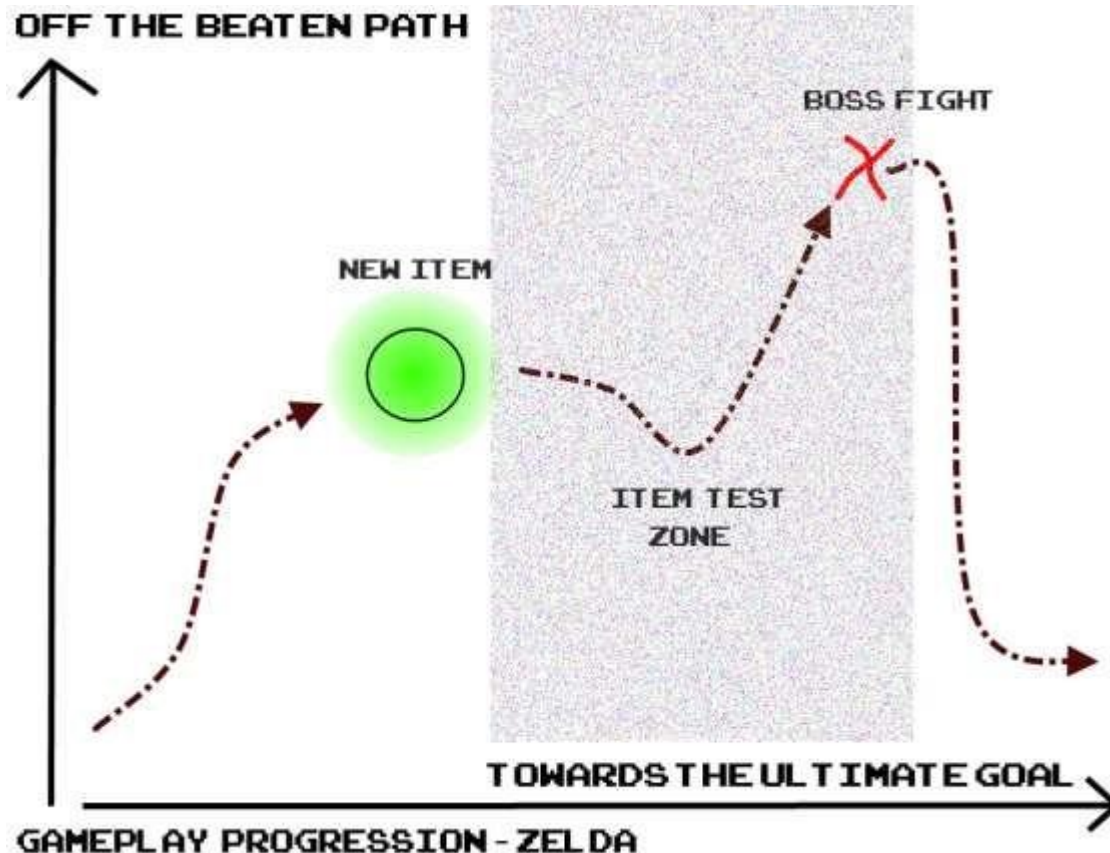
Provides Useful Mastery

- No level design, no mastery



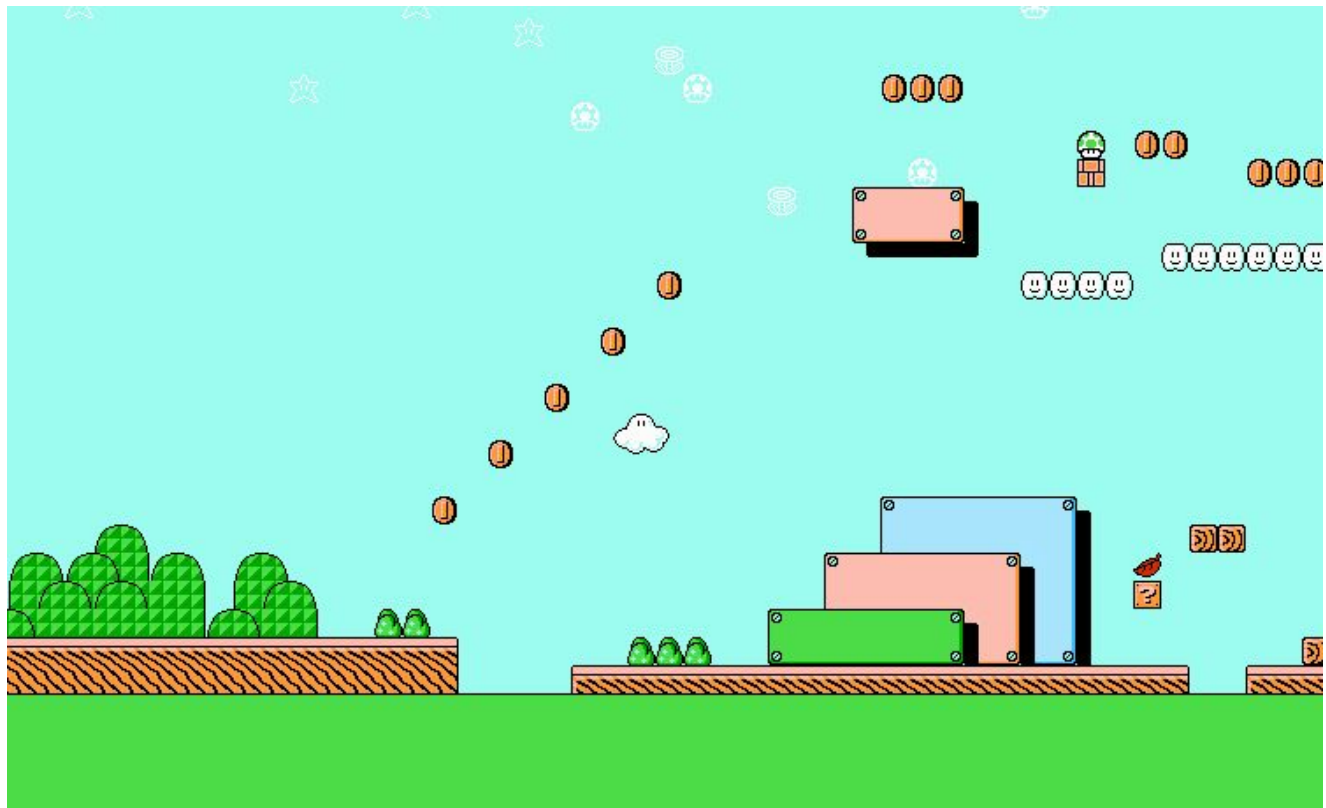
Provides Useful Mastery

- Enter a dungeon -> Get a weapon -> Test the weapon



Influences Player's Decision Making

- Nintendo uses coins to tell that there's a secret path above.



Teaches Player About Game's Mechanics

- Super Mario Bros. (NES)
 - Powerup
 - Powerup physics
 - Jumping Mechanics
 - Secrets (Scenario based)



Teaches Player About Game's Mechanics

- The Legend of Zelda: The Minish Cap
- Link pulls a lever to gain access to a mushroom



Teaches Player About Game's Mechanics

You can also pull the mushroom as the dirt is also placed near the mushroom (hint)



Teaches Player About Game's Mechanics

- How far you pull the mushroom affects how far you jump. (see the gap between dirts)



Teaches Player About Game's Mechanics

- New Weapon : Gust Jar
- Used by Link to suck things. He can't escape this room unless he understands how to use Gust Jar.



Teaches Player About Game's Mechanics

- It can also be used to shoot gust of wind. Again, Link can't progress unless he has mastered this mechanic.



Teaches Player About Game's Mechanics

- You can also do this!



Good Level Design?

- Reduces amount of tutorials
- Balance between difference in scale and difference in kind
- Driven by core mechanics



Difference in Scale

- Scales a variable from game's mechanics (enemy's hp, game speed, number of blocks, etc).
- Easy way to provide variety.

Difference In Scale

- Falling blocks speed gets faster as you progress



Difference In Scale

Monster AAA

- HP : 50
- STR : 5
- DEF : 3



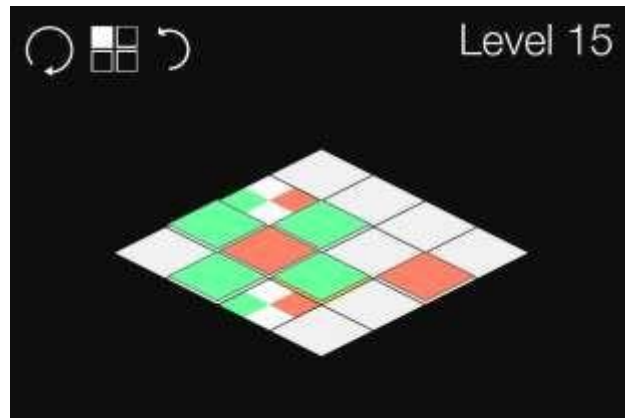
Monster BBB

- HP : 150
- STR : 13
- DEF : 27

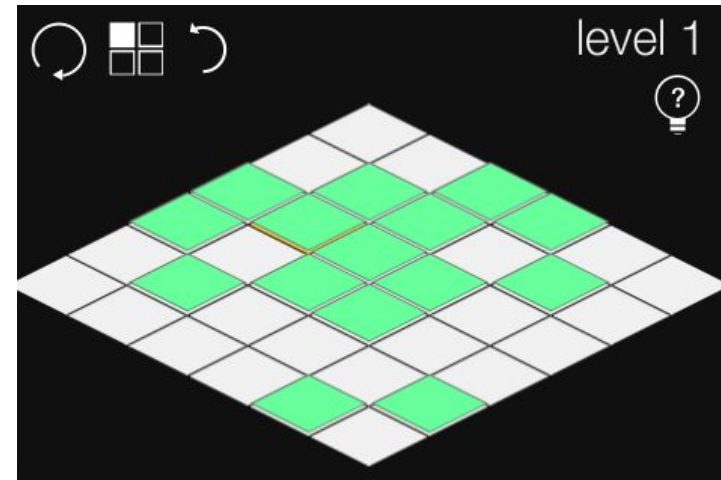


Difference In Scale

- 4x4



- 6x6

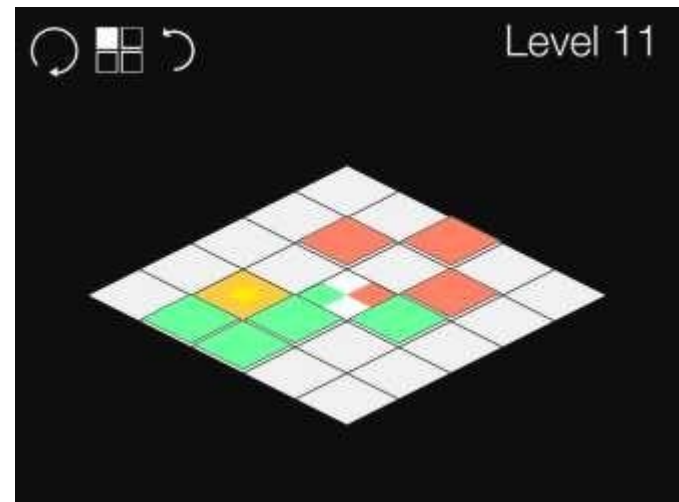
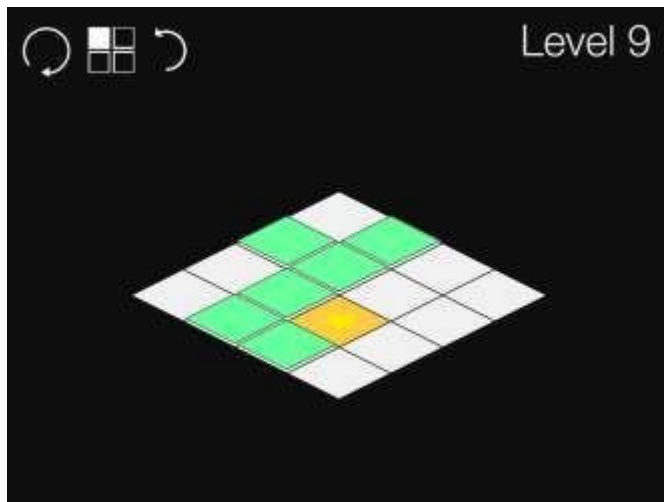


Difference In Kind

- Variety in Behavior
- Encourages or Challenges player to think differently.

Difference in Kind

- Basic tiles
- Color Switching tiles



Driven by Core Mechanics

Mario



Megaman



Castlevania



Contra



Speed	Med. Accel	Binary	Binary	Binary
Jump	Steerable	Cancellable	Fixed	Steerable
Weapon	Jump Fireball	Bullet Powerup	Whip Sub-weapon	8 dir shot, Powerup

Driven by Core Mechanics

- Mario level design requires more precise platforming



tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

Kishōtenketsu

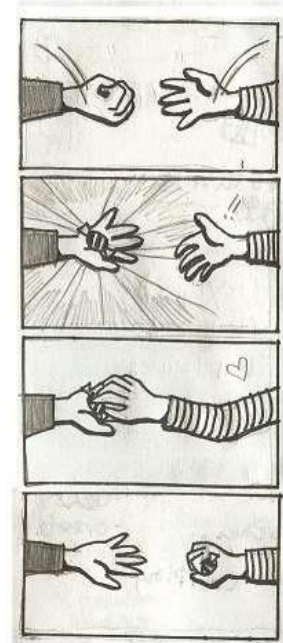
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Four Step Level Design

Four Step Level Design

- Used by Koichi Hayashida (Nintendo EAD) to design levels in Mario 3D Land and 3D World.
- Inspired by narrative structure of four-panel comics
- 1st panel Introduction
- 2nd panel Development
- 3rd panel Twist
- 4th panel Conclusion



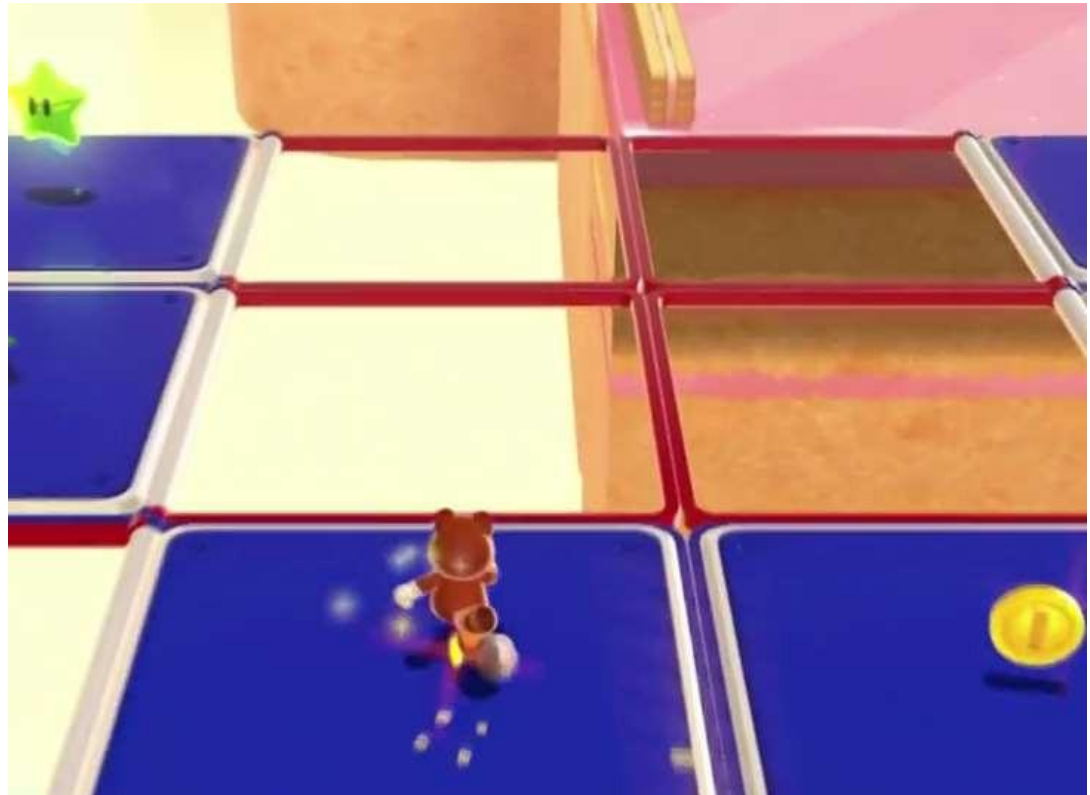
Four Step Level Design

- Introduction
- Player are given a safe place to learn the concept of this level



Four Step Level Design

- Development
- You fail, you die



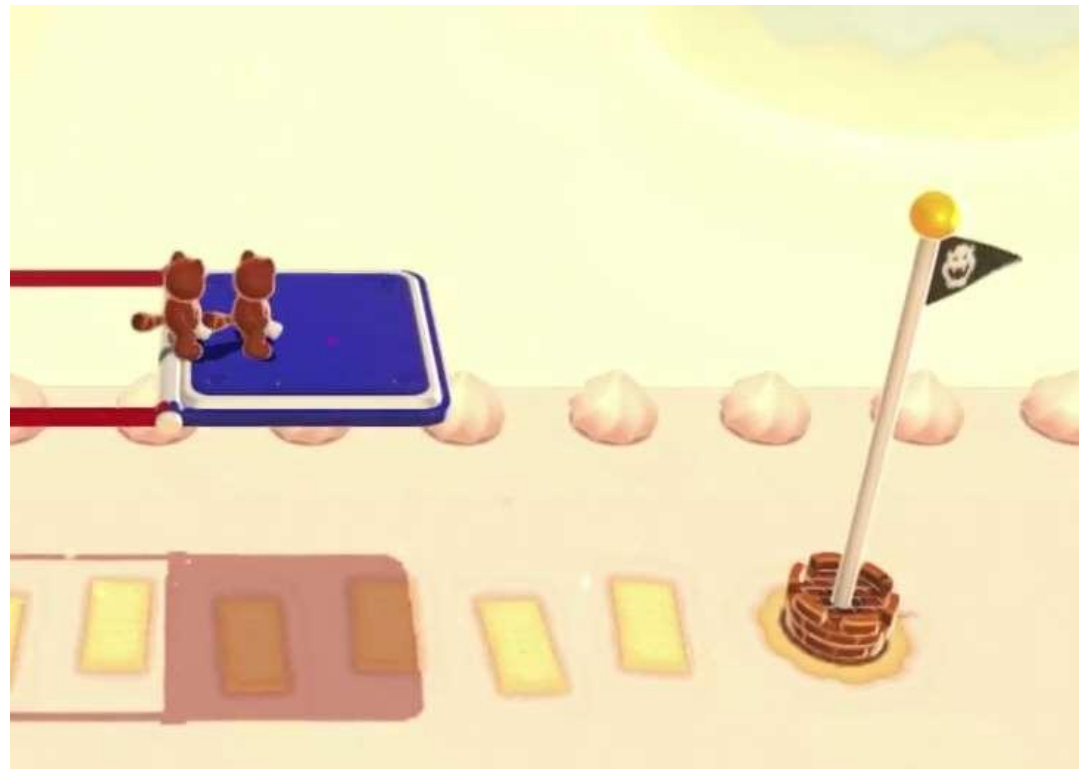
Four Step Level Design

- Twist
- Add twist to the concept that surprises players



Four Step Level Design

- Conclusion
- Give player one last chance to show off what they've learned to end the level.



Miyamoto on World 1-1

[Miyamoto on World 1-1: How Nintendo made Mario's most iconic level](#)



Demo (level editor use cases)

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
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spec. PSI

Portal 2 [\[#1\] Minecraft + Portal 2 - Escape - Vertez](#)

Pullblox (3DS)

[smwcentral.net](#) → [FIXING the ENTIRE SM64 Source Code \(INSANE N64 performance\)](#)

Side note: Fallout 3 mods - <https://youtu.be/gWVkh-0oQzw?t=51s>, <http://www.nexusmods.com/fallout3/mods/categories/>

[trle.net](#) - https://en.wikipedia.org/wiki/Tomb_Raider_Chronicles

For the PC version, Core Design released the level building tools as a **level editor on a second disc**. This was done as Chronicles would be the **last game using that generation of technology**, and they wanted to **allow fans the freedom to create levels** of their own

[https://en.wikipedia.org/wiki/Donkey_Kong_\(Game_Boy\)](https://en.wikipedia.org/wiki/Donkey_Kong_(Game_Boy))

Ten years later (in 2004), an enhanced version of the Game Boy title for the Game Boy Advance was planned, titled **Donkey Kong Plus**. In addition to featuring enhanced graphics and backgrounds, the proposed remake also featured a **level designer accessible through the GameCube**. The game ultimately resurfaced as **Mario vs. Donkey Kong**, a completely new game with similar gameplay. It was followed by a sequel titled **Mario vs. Donkey Kong 2: March of the Minis**, which featured a level designer.

[https://pl.wikipedia.org/wiki/Janusz_Pelc_\(programista\)#Gry](https://pl.wikipedia.org/wiki/Janusz_Pelc_(programista)#Gry)

Full Screen Mario (HTML) <https://web.archive.org/web/20160429152213/https://github.com/FullScreenShenanigans/FullScreenMario/archive/master.zip>

Super Mario Maker

https://www.youtube.com/watch?v=4lFtKF_VQfc

[Super Mario Maker - 19 Amazing Automatic Levels](#)